

DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 211: THEORY OF STATISTICS

Session: 2025-26

Semester: (II)

Quiz 2

Maximum Marks: 20

Time: 50 minutes

1. Let X_1, X_2, X_3, X_4 be a random sample from a Poisson distribution with parameter $\lambda > 0$. Apply the NP lemma to derive the Uniformly Most Powerful (UMP) test of exact size $\alpha = 0.05$ for testing $H_0 : \lambda = 1$ vs. $H_1 : \lambda > 1$.
Verify whether this test is UMP for testing $H_0 : \lambda \leq 1$ vs. $H_1 : \lambda > 1$, or not.

Given: If $Y \sim \text{Poisson}(4)$, then $P(Y \geq 7) = 0.1107$, $P(Y \geq 8) = 0.0511$, $P(Y \geq 9) = 0.0214$, and $P(Y \geq 10) = 0.0081$. **(7 marks)**

2. Let $X_1, X_2, \dots, X_n$ be a random sample from a Rayleigh distribution with probability density function:

$$f(x; \sigma) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)}, \quad x > 0, \sigma > 0$$

Show that the family has MLR property in some statistic $T(X_1, X_2, \dots, X_n)$.

Hence, construct the UMP size- α test for $H_0 : \sigma = \sigma_0$ against $H_1 : \sigma > \sigma_0$.

Obtain the distribution of this statistic and express the critical region in terms of cut-off values (in the standard form) of this distribution. **(7 marks)**

3. Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ distribution, where σ^2 is known. Construct the small sample exact Likelihood Ratio test of size α for:

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_A : \mu < \mu_0$$

Derive the test statistic and express the critical region in its simplest form.

(6 marks)